Sensitivity of Continuous Performance Test (CPT) at Age 14 Years to Developmental Methylmercury Exposure

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Hit Reaction Time latencies (HRT) in the Continuous Performance Test (CPT) measure the speed of visual information processing. The latencies may involve different neuropsychological functions depending on the time from test initiation, i.e., first orientation, learning and habituation, then cognitive processing and focused attention, and finally sustained attention as the dominant demand. Prenatal methylmercury exposure is associated with increased reaction time (RT) latencies. We therefore examined the association of methylmercury exposure with the average HRT at age 14 years at three different time intervals after test initiation. A total of 878 adolescents (87% of birth cohort members) completed the CPT. The RT latencies were recorded for 10 minutes, with visual targets presented at 1000 ms intervals. After confounder adjustment, regression coefficients showed that CPT-RT outcomes differed in their associations with exposure biomarkers of prenatal methylmercury exposure: During the first two minutes, the average HRT was weakly associated with methylmercury (beta (SE) for a ten-fold increase in exposure, (3.41 (2.06)), was strongly for the 3-to-6 minute interval (6.10 (2.18)), and the strongest during 7–10 minutes after test initiation (7.64 (2.39)). This pattern was unchanged when simple reaction time and finger tapping speed were included in the models as covariates. Postnatal methylmercury exposures did not affect the outcomes. Thus, these findings suggest that sustained attention as a neuropsychological domain is particularly vulnerable to developmental methylmercury exposure, indicating probable underlying dysfunction of the frontal lobes. When using CPT data as a possible measure of neurotoxicity, test results should therefore be analyzed in regard to time from test initiation and not as overall average reaction times.